Fractions: Concept & Equivalence

Year 4 · **Homework Worksheet**

Worked Example

Worked example: Simplify 6/9.

Answer: The highest common factor of 6 and 9 is 3. 6÷3=2, 9÷3=3, so 6/9 = 2/3.

Easy / Standard (15 questions)

1. In the fraction 4/5, which number is the numerator and which is the denominator?
2. Which of these is larger? A) 3/6 B) 4/6 C) They are equal D) Cannot tell
3. Shade 1/2 of a bar with 2 equal parts. What fraction is NOT shaded?
4. What is 1/10 of 70?
5. Which fraction is equivalent to 1/5 with a denominator of 20? A) 2/20 B) 4/20 C) 5/20 D) 20/5
6. In the fraction 1/3, which number is the numerator and which is the denominator?
7. Which is larger: 2/6 or 3/6?
8. A bar has 4 equal parts and 1/4 is shaded. Which fraction is NOT shaded? A) 1/4 B) 2/4 C) 3/4 D) 4/4
9. What is 1/5 of 25?
10. Write an equivalent fraction to 1/2 with a denominator of 4.
11. In the fraction 1/4, what is the numerator? A) 1 B) 4 C) 3 D) 14
12. Which is larger: 4/8 or 5/8?
13. Shade 3/5 of a bar with 5 equal parts. What fraction is NOT shaded?
14. Select ALL the fractions below that are equivalent to 1/5: A) 2/10 B) 3/15 C) 4/25 D) 5/20
15. Write an equivalent fraction to 1/5 with a denominator of 15.

Standard (15 questions)

1. What is 3/5 of 10?
2. Simplify 5/10 to its lowest terms. A) 1/5 B) 1/2 C) 2/5 D) 5/10
3. Convert the improper fraction 15/4 to a mixed number.
4. 3/5 + 1/5
5. 3/5 − 2/5 = ? A) 1/5 B) 1/10 C) 5/10 D) 1
6. What is 5/6 of 24?
7. Simplify 2/4 to its lowest terms.
8. Convert the improper fraction 10/3 to a mixed number. A) 3 1/3 B) 3 1/2 C) 2 1/3 D) 3 2/3
9. 8/10 + 1/10
10. 3/10 − 1/10
11. What is 3/8 of 24? A) 3 B) 6 C) 9 D) 12

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12. Simplify 3/6 to its lowest terms.

13. Convert the improper fraction 5/2 to a mixed number.

14. Select ALL the fractions below that simplify to 1/2: A) 5/10 B) 3/9 C) 4/8 D) 2/6

15. 1/10 + 8/10

Standard / Advanced (15 questions)

1. Convert 9/4 to a mixed number, then add 1 whole.

2. 4/10 + 3/10 − 2/10

3. Find 1/3 of 18, then double it. A) 6 B) 9 C) 12 D) 18

4. Simplify 3/6, then convert your answer to a fraction with denominator 12.

5. Which is larger: 2/3 or 5/9? Convert 2/3 to a fraction with denominator 9 first.

6. In a school race with 24 runners, 1/4 of them finish in under 5 minutes. Of the runners who do NOT finish under 5 minutes, half finish in the next minute. How many runners finish in the next minute?

7. A jug holds 10 litres when full. 1/5 of the water is poured out to water plants, then 2 litres is poured back in. How much water is in the jug now? A) 8 litres B) 9 litres C) 10 litres D) 11 litres

8. Ava spends 1/4 of £40 on a book and 1/8 of £40 on a magazine. How much money is left?

9. A box contains 24 chocolates. 1/4 are given to a class on Monday. The next day, half of what remains is given away too. How many chocolates are left after Tuesday?

10. A cake is cut into 16 equal pieces. At a party, 3/4 of the pieces are eaten, and the rest are shared equally between 2 friends. Which of the statements below are TRUE? Select ALL that apply. A) 12 pieces were eaten B) Each friend receives 2 pieces C) Each friend receives 4 pieces D) 1/4 of the cake remains

11. A jug holds 12 litres when full. 1/3 of the water is poured out to water plants, then 3 litres is poured back in. How much water is in the jug now?

12. Jack spends 1/5 of £30 on a book and 1/10 of £30 on a magazine. How much money is left? A) £18 B) £20 C) £21 D) £24

13. A box contains 18 chocolates. 1/3 are given to a class on Monday. The next day, half of what remains is given away too. How many chocolates are left after Tuesday?

14. In a school race with 16 runners, 1/4 of them finish in under 5 minutes. Of the runners who do NOT finish under 5 minutes, half finish in the next minute. How many runners finish in the next minute?

15. A cake is cut into 20 equal pieces. 2/5 of the pieces are eaten, and the rest are shared equally between 4 friends. Which of the statements below are TRUE? Select ALL that apply. A) 8 pieces were eaten B) Each friend receives 3 pieces C) Each friend receives 4 pieces D) 1/2 of the cake remains

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